

Signal Processing

Exercise 1 – Fourier Series

Summer Semester 2026

Exercise 1

The periodic signal $f(t)$ is defined as

$$f(t) = \begin{cases} 1, & \text{for } -\frac{T_0}{3} \leq t < +\frac{T_0}{6} \\ 0, & \text{for } +\frac{T_0}{6} \leq t < +\frac{2}{3}T_0 \end{cases} ; \quad f(t) = f(t + n \cdot T_0), \quad n = \pm 1, \pm 2, \dots$$

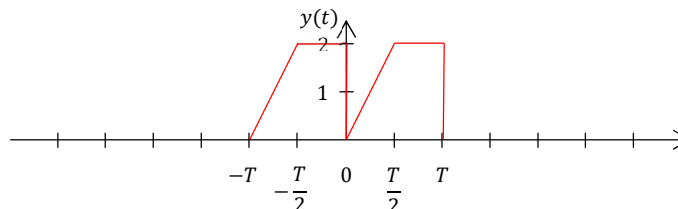
- Sketch the signal $f(t)$ for $t = -\frac{4}{3}T_0$ to $t = \frac{7}{6}T_0$.
- Calculate the real and complex Fourier coefficients.
- Sketch the line spectrum (magnitude spectrum) of $|c_n|$ for $n \leq 5$.

Exercise 2

We will consider the signal $y(t) = \cos\left(\omega_0 \cdot \left(t - \frac{T_0}{3}\right)\right)$. Calculate the real and complex Fourier coefficients.

Exercise 3

The periodic signal $y(t)$ is given by the following sketch.



- Calculate the coefficient a_0 of the DC component.
- Derive a formula which allows the calculation of each coefficient a_n of the even signal component.
- Derive a formula which allows the calculation of each coefficient b_n of the odd signal component.

Exercise 4

Let $y(t)$ be a periodic signal. Describe how the Fourier coefficients of the following variations $y_{\text{var}}(t)$ can be derived from the Fourier coefficients of $y(t)$.

a) $y_{\text{var}}(t) = y(t) + 4$

b) $y_{\text{var}}(t) = 2 \cdot y(t)$

c) $y_{\text{var}}(t) = y(3t)$